

FIT3152 Data analytics – S1 2026: Assignment 2

Your task	- The objective of this assignment is to gain familiarity with classification models using R. - This is an individual assignment.
Value	- This assignment is worth 20% of your total marks for the unit. - It has 40 marks in total.
Suggested Length	8 – 10 A4 pages, approximately 1,000 words (for your report) + extra pages as appendix for your R script. - Font size 11 or 12pt, single spacing.
Due Date	11.55pm Friday 15th May 2026
Submission	- Submit a single PDF file and single video presentation file on Moodle. - Use the naming convention: <i>FirstnameSecondnameID.{pdf, mp4, mov etc.}</i> - Turnitin will be used for similarity checking of all written submissions.
Generative AI Use	Statement on Generative AI required by the university: In this assessment, you can use generative artificial intelligence (AI) in order to search for R functions and examples to perform tasks that you specify only. Any use of generative AI must be appropriately acknowledged (see Learn HQ).
Late Penalties	5% (2 mark) deduction per calendar day for up to one week. - Submissions more than 7 calendar days after the due date will receive a mark of zero (0) and no assessment feedback will be provided.

Instructions and data

The objective of this assignment is to gain familiarity with classification models using R. You will be using a modified version of the World Values Survey (WVS) data that was used for Assignment 1. We now want to create Machine Learning models to predict the level of confidence in different social organizations based on participant responses to other selected attributes. For Assignment 2, confidence in individual social organisations has been re-coded as binary class variables. You can read more on the World Values Survey here: <https://www.worldvaluessurvey.org/WVSContents.jsp>

There are two options for compiling your written report:

- (1) You can create your report using any word processor with your R code pasted in as machine-readable text as an appendix, and save as a pdf, or
- (2) As an R Markdown document that contains the R code with the discussion/text interleaved. Render this as an HTML file and save as a pdf.

Your video report should be less than 100MB in size. You may need to reduce the resolution of your original recording to achieve this. Use a standard file format such as .mp4, or mov for submission.

Creating your data set

Clear your workspace, set the number of significant digits to a sensible value. Download “WVSBinaryExtract.csv” and create your individual data using the following code. Your individual data will have Country, Wave, **three** different binary class variables (confidence in social organisations) and **30** coded participant responses (attributes), with **20,000** observations.

```
rm(list = ls())
set.seed(XXXXXXX) # Your Student ID is the random seed
WD = read.csv("WVSBinaryExtract.csv")
selected_cols = c(sample(3:49, 30), sample(50:63, 3))
WD = WD[c(1:2, selected_cols)]
WD = WD[sample(nrow(WD), 20000, replace = FALSE),]
```

Use your individual data created above to answer all questions.

Questions (18 Marks)

1. Explore the data: What is the proportion of “**High**” (Confidence = 1), to “**Low**” (Confidence = 0) for your **three binary class variables**? Obtain descriptions of the other predictor attributes. Is there anything noteworthy in the data? Are there any attributes you need to consider **omitting from your analysis**? **(2 Marks)**

all**

2. Document any **pre-processing** required to make the data set suitable for the model fitting that follows. **(1 Mark)**

(last year no mark for pre-processing)

3. Divide your data into a 70% training and 30% test set by adapting the following code (written for the iris data). Use your student ID as the random seed.

```
set.seed(XXXXXXX) #Student ID as random seed
train.row = sample(1:nrow(iris), 0.7*nrow(iris))
iris.train = iris[train.row,]
iris.test = iris[-train.row,]
```

4. Using your training data and each of the techniques below, implement models to separately classify **each** of your **three class variables**. For this question you may use the R functions at their default settings if suitable. Note: **Do not use any class variables as predictor attributes**. **(5 Marks)**

- **Decision Tree**
- **Naïve Bayes**
- **Bagging**
- **Boosting**
- **Random Forest**

ask the program to 3 times by themselves, not write ourselves

can be found in applied session & notes

ensemble method (red; 3)

more towards interpretation, academic writing, instead of R code

30%

5. Using the **test data**, classify each of the test cases as “High” (Confidence = 1) or “Low” (Confidence = 0) for each of your three class variables using each of the models implemented in Question 4. Create **confusion matrices** and report the **accuracy, precision, recall, and F1-score** of each model in classifying each class variable. Summarise your results in **three tables**, one for each class variable. **(3 Marks)** label your table, produce table 3 times, provide confusion matrix

6. Using the **test data**, calculate the **confidence** (probability) of predicting “High” (Confidence = 1) or “Low” (Confidence = 0) using **each classifier**, and construct **ROC curves** for **each** of your **three class variables**. Make **a separate plot** for each class variable with all the curves for that class on the same axis. Use a different colour for each classifier. This will give you **three plots with five curves in each**. Calculate and report the AUC for each classifier over each class variable. **(4 Marks)**

senior 1, we 3

pls compare which is the best model for every questions

applied session*, interpretation (3 plots), all the numbers

7. Compare the results in Questions 5 and 6 in relation to the **performance of all classifiers** across your **three class variables**. From the five classifiers, is there a **single “best” classifier** for **predicting each of the classes**? Is there a **“worst” ensemble classifier overall**? Do the **“best” or “worst” classifiers** vary between your **three class variables**? Explain your reasoning. **(3 Marks)**

compare the ensemble in Q4 (only 3)

Figure 1. The bar charts for ... (a) bagging (b) boosting and (c) random forest

Investigative Tasks (18 Marks)

put 3 diagrams together


(a) (b) (c)

8. Examining each of the **ensemble models** for each of your **three class variables**, determine the **most important attributes** in predicting confidence. Summarise your results in **three tables** for each class variable. Which **attributes** are important overall for predicting confidence in **each class variable**? Which **attributes** could be omitted from the data with very **little effect** on performance? **Give reasons. (5 Marks)**

will be repeated 3 times, not one table only (senior one set only)

9. Looking at the relative strengths and weaknesses of each model, and consulting relevant **references**, can you explain **any difference in performance between the models** across each of the **class variables**? That is, **why do some of these models perform better** than others on the type of data to be fitted? **(2 Marks)**

Those are examples, in Q9 guidelines, (more toward explanation), adapt with datasets

- ~~10.~~ Improve the **worst performing ensemble** model from Question 7. You may do this by **adjusting parameters, attribute selection, cross-validation** of the basic ensemble model and/or any other applicable process. Show that your **new model is better using F1-scores, ROC curves and AUC**. Describe how you created your improved model. What factors were important in your decision? State why you chose the attributes you used. **(4 Marks)**

Try to improve your model, write yourself

11. Using the insights from your analysis so far, implement an **Artificial Neural Network** classifier to compare predictability **over time** for the country with the greatest number of observations in your data. Using the training data for that country, fit an Artificial Neural Network model to classify confidence in **one of your three class variables**. Describe the attributes used and **data pre-processing** for implementation. **(3 Marks)**

(time is important, idea for preprocessing)

write yourself, not using tree, using ensemble

wave*

X12.

Using the Artificial Neural Network model from Question 11, classify test instances of the **selected country** separately **for the two Waves** having the greatest number of observations in your test data. How does the model performance **change over the two waves**? Use F1-scores, ROC curves and AUC to compare. **(4 Marks)**

totally different with senior, not related X10, X12

how you identify, graphical method, why selected? (evidence & justification; box plots), plot 1 diagram with all the countries with the highest observations, show all evidence interpretation*

Report and Video Presentation (4 Marks)

Write a brief report (suggested length 8 – 10 pages) summarizing your results. Include your R script, copied and pasted as machine readable text, as an appendix. Use commenting in your R script, where appropriate, to help a reader understand your code. Alternatively combine working, comments and reporting in R Markdown. **(2 Marks)**

Record a short presentation using your smart phone, Zoom, or similar method. Your presentation should be approximately 5 minutes in length and summarise your main findings, as well as describing how you conducted your research, and any assumptions made. Pay particular emphasis to your results for the investigative tasks. **(Submission Hurdle and 2 Marks)**

Software

It is expected that you will use R for your data analysis and graphics and tables. You are free to use any R packages you need but please document these in your report and include in your R code.

Description of the data

Most fields are on integer scales over varying ranges. The convention is that larger numbers generally indicate greater agreement with statement or frequency of occurrence. Fields in bold at the end indicate binary class variables giving the level of confidence each participant has on the relevant social organization. “High confidence” (Confidence = 1), to “Low confidence” (Confidence = 0).

You can access more detail on each field in the WVS-7 Master Questionnaire 2017-2020 English.pdf, linked from <https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp> Use the question ID given in the **WVS Wave 7 Reference** in the table below.

Column Name	Original Descriptor	WVS Wave 7 Reference
Country	ISO 3166-1 alpha-3 country code	B_COUNTRY_ALPHA
Wave	Chronology of EVS-WVS waves	A_WAVE
ILFam	Important in life: Family	Q1
ILFriends	Important in life: Friends	Q2
ILLeisure	Important in life: Leisure time	Q3
ILPolitics	Important in life: Politics	Q4

ILWork	Important in life: Work	Q5
ILReligion	Important in life: Religion	Q6
Happy	Feeling of happiness	Q46
Health	State of health (subjective)	Q47
ICQIndependence	Important child qualities: independence	Q8
ICQHardWork	Important child qualities: Hard work	Q9
ICQResonsibility	Important child qualities: feeling of responsibility	Q10
ICQImagination	Important child qualities: imagination	Q11
ICQTolerance	Important child qualities: tolerance and respect for other people	Q12
ICQThrift	Important child qualities: thrift saving money and things	Q13
ICQDetermination	Important child qualities: determination perseverance	Q14
ICQFaith	Important child qualities: religious faith	Q15
ICQUnselfishness	Important child qualities: unselfishness	Q16
ICQObedience	Important child qualities: obedience	Q17
ACTReligion	Active/Inactive membership of church or religious organization	Q94
ACTSport	Active/Inactive membership of sport or recreation	Q95
ACTArtsEd	Active/Inactive membership of art, music, educational	Q96
ACTUnions	Active/Inactive membership of labour unions	Q97
ACTPolParty	Active/Inactive membership of political party	Q98
ACTEnvOrg	Active/Inactive membership of environmental organization	Q99
ACTProfessional	Active/Inactive membership of professional organization	Q100
ACTHumanitarian	Active/Inactive membership of charitable/humanitarian organization	Q101
Happy	Feeling of happiness	Q46
Health	State of health (subjective)	Q47
Trusted	Most people can be trusted	Q57
LifeSatis	Satisfaction with your life	Q49
FutureWork	Future changes: Less importance placed on work	Q43
FutureTech	Future changes: More emphasis on technology	Q44
FutureRespect	Future changes: Greater respect for authority	Q45
PolInterest	Interest in politics	Q199
PolPetition	Political action: Signing a petition	Q209
PolBoycott	Political action: joining in boycotts	Q210
PolDemons	Political action: attending lawful/peaceful demonstrations	Q211
PolScale	Self positioning in political scale	Q240
IncomeEquality	Income equality	Q106
PrivateState	Private vs state ownership of business	Q107
GovtRespons	Government responsibility	Q108
Competition	Competition good or harmful	Q109
HardWork	Hard work brings success	Q110

PolLeader	Political system: Having a strong leader	Q235
PolExperts	Political system: Having experts make decisions	Q236
PolArmy	Political system: Having the army rule	Q237
PolDemoc	Political system: Having a democratic political system	Q238
Sex	Sex	Q260
Age	Age	Q262
CChurches	Confidence: Churches	Q64
CArmedForces	Confidence: Armed Forces	Q65
CPress	Confidence: The Press	Q66
CUnions	Confidence: Labour Unions	Q68
CPolice	Confidence: The Police	Q69
CParliament	Confidence: Parliament	Q73
CCivilService	Confidence: The Civil Services	Q74
CTelevision	Confidence: Television	Q67
CGovernment	Confidence: The Government	Q71
CPolParties	Confidence: The Political Parties	Q72
CMajComp	Confidence: Major Companies	Q77
CEnvProt	Confidence: Environmental Protection Movement	Q79
CWomensMvt	Confidence: The Womens' Movement	Q80
CCourts	Confidence: Justice System/Courts	Q70